



INFLUENCE OF GEOPOLITICAL POSITION ON NEWS MEDIA NARRATIVES

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Abstract

This thesis explores how natural language processing (NLP) tools can uncover if and how geopolitical position impacts the framing of news media narratives covering the same global event. Thematic choices and emotional tone are examined in how they differ on a global scale. Topic modeling with Latent Dirichlet Allocation (LDA) accompanied by Principal Component Analysis (PCA) for topic distributions and sentiment analysis with a baseline Valence Aware Dictionary and sEntiment Reasoner (VADER) and state-of-the-art Robustly Optimized BERT Pretraining Approach (RoBERTa) model approach have been selected to investigate this. The dataset this methodology is applied to contains 1,674 articles collected from 47 outlets across four regional groups covering the 2025-26 Greenland crisis between the period of January 2025 to February 2026. Groups consist of three stakeholder groups: Greenland & Denmark, affected European countries and the United States of America, and one unaffiliated group. Results suggest a relatively consistent narrative of sovereignty and national security possessed by the primary threatened stakeholder; Greenland & Denmark. Group Other appears to hold the most diverse use of vocabulary when narrating under the same thematic labels. Furthermore, USA scored the least positively in sentiment while affected Europe the most positive. The remaining two regions share similar emotional tones, captured by RoBERTa that differs substantially from the almost level-headed VADER results. These findings demonstrate an influence of global position on the thematic framing and emotional tone exhibited in news media and the ability for state-of-the-art NLP tools to bring forth these subtle differences.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

The dataset is owned and assembled by the author of this thesis. Work on this thesis did not involve collecting data from human participants or ani-

mals. The author of this thesis retains ownership of the data and code during and after the completion of this thesis. The dataset consists of publicly published article texts from established media outlets, acquired through article URL retrieval via the GNews Python library (v0.3.x) from Google News. Using the newspaper3k library, article body texts were extracted depending on the accessibility of article websites. All figures and tables in this thesis were created by the author with Python. Data collection, pre-processing and analysis code was written by the author in Python 3.11, with libraries and modules: GNews, googlenewsdecoder, newspaper3k, pandas, transformers (v4.40.0), torch (v2.2.2), scikit-learn (v1.x), matplotlib (v3.x), seaborn (v0.13.x), scipy (v1.x), wordcloud (v1.x), and tqdm. The thesis code can be accessed through the GitHub repository following the link <https://github.com/matholdeman/bachelor-thesis.git>. The adapted code fragments are clearly indicated in the notebook. A generative language model Claude under Anthropic (2025) was used to improve the author's original code. Grammarly Inc. (2025) was used for spell checking and grammar. Overleaf with (L^AT_EX) was used for typesetting.

2 INTRODUCTION

Media plays a primary role in informing the public of major political and socioeconomic events in the world, with news media serving as a pivotal source of information worldwide. Consequently, media outlets have significant influence in shaping the opinions of a nation's individuals, groups and societies as a whole. This thesis explores the narratives employed by different, widely reached news media outlets based on a particular event to analyze the dominating themes and sentiment. Grouping articles based on regional position, the difference in media narratives will be determined to infer the subtle biases news outlets pertain for geopolitical reasons. The 2025-26 Greenland crisis is therefore considered as this subject has received a considerable amount of global media attention since U.S. President Donald Trump publicly expressed his interest in buying the island.

Due to the massive spread of information online, it is often difficult to decipher the 'truthfulness' of sources of media. While news outlets are generally portrayed as unbiased, the possible presence of underlying nuances is often overlooked when consuming such media. Missing such subtleties can have severe repercussions as it can sway the audience to believe something that doesn't necessarily reflect the ground truth.

Hence, the following research question will tackle this matter:

To what extent does geopolitical alignment influence the narratives of news media outlets covering the 2025-26 Greenland crisis?

To further investigate this research question, the following sub-questions will be addressed:

- RQ1 Which themes dominate the framing of the crisis in the media of the regions involved?
- RQ2 How does the emotional tone in news media differ between the regions involved?
- RQ3 To what extent does news media based in unaffiliated regions of the crisis align with or diverge from the narratives and tone of affected regions?

3 RELATED WORK

The presence of bias in news media is a reoccurring theme across research in politics. [Davis, Worsnop, and Hand \(2022\)](#) identified significant gender bias in popular U.S. media outlets; words describing male politicians were predominantly associated with higher ranking government positions while negative sentiment was attributed to female politicians. This demonstrates the broader capacity of formal media to hide subtle biases in favor of certain political ideologies.

A regional comparison study conducted by [Wang and Yin \(2022\)](#) found that local vs non-local newspapers shared similar framing choices under the subject of the Shanghai COVID-19 outbreak in 2022 through Latent Dirichlet Allocation modeling. However, faint thematic differences were uncovered, with local newspapers focusing on impact on the stock market while non-local newspapers explored its effect on citizens. Additionally, [Albalawi, Yeap, and Benyoucef \(2020\)](#) comparatively evaluated different topic modeling techniques on social media data, indicating that at smaller numbers of topics, LDA produced more interpretable topic clusters than other methods. These findings are directly relevant to this thesis as it confirms the abilities of LDA to identify coherent thematic clusters and explore regional differences.

The focal point in recent research has shifted from manual content analysis to computational natural language processing (NLP) methods, extending the scope of media bias research. [Griswold, Robbins, and Pollard \(2025\)](#) deduced that lexicon-based models such as VADER under-perform severely in distinguishing stances in corpora and recommends the usage and fine-tuning more advanced methodologies. [Nathan, Jacknius, Atha](#)

Rabbaanii, Hasani, and Maulina (2025) applied a fine-tuned RoBERTa model to investigate political bias, capturing fluctuations in sentiment across the 2024 U.S. presidential elections. Similarly, Gamea, Hasan, Serour, Rezk, and Gomaa (2026) transitioned from a baseline VADER to advanced RoBERTa model to unravel sentiment in The Guardian’s coverage of the Israeli-Palestinian conflict. Findings in both papers suggest that sentiment is a justifiable indicator of media narratives and that transformer-based models outperformed traditional models in detecting sentiment. Therefore, these approaches are adopted in this thesis.

Several gaps in research exists to motivate the pursuit of this thesis. Firstly, current studies are bounded to coverage from stakeholder or single-country news media, disregarding the framing choices of uninvolved parties. Secondly, prior research focuses primarily on a single path to explore narratives. This thesis combines both topic modeling and sentiment analysis to extend such explorations. Lastly, the thematic and tonal choices in the framing of the 2025-26 Greenland crisis by the media has not vastly been explored. This thesis addresses all three gaps by harnessing an NLP pipeline on a cross-regional corpus.

4 METHODS

Articles were collected with a built scraper, using the Google News (GNews) Python library along with the googlenewsdecoder package to acquire article URLs linked to its original source address. 67 total media outlets were targeted based on audience reach and English-language availability across four regional groups, distinguished based on impact on the crisis:

- Group 1 (directly threatened stakeholder): Greenland & Denmark
- Group 2 (tariff-threatened): Affected Europe (The Netherlands, France, Sweden, Norway, Finland, Germany & United Kingdom)
- Group 3 (opposing party to the previous two groups): United States of America
- Group 4 (unaffiliated regions chosen based on geopolitical position to the crisis, cultural diversity and availability of English articles): Other (Canada, India & Australia)

4.1 *Mitigating selection bias*

Considering the impact of political affiliation to skewing results, outlets in the United States regional group were selected based on the [AllSides](#)

(n.d.)’s Media Bias Chart to maintain ideological diversity, aiming for two outlets per political affiliation:

1. 1. Left: New York Times, AP News
2. 2. Left leaning: CNN, Washington Post
3. 3. Center: News Week, Straight Arrow News
4. 4. Right leaning: New York Post, Washington Examiner
5. 5. Right: Fox News, The Spectator

The selection of media outlets under the affected Europe, Greenland & Denmark and the Other regional groups were primarily focused on the Reuters’ Digital News Report [Newman, Ross Arguedas, Robertson, Nielsen, and Fletcher \(2025\)](#) that identifies news media outlets with the highest % weekly reach for a variety of countries. However, due to most highest-reach outlets publishing exclusively in their native languages, many outlets were selected for their availability of English articles, which was often limited.

4.2 3-stage verification process

Outlets were queried with search term "Greenland" within the timeframe of January 1, 2025 to February 28, 2026 to account for the most critical period within the crisis. To avoid Google’s IP blocking infrastructure, a delay between 2-4 seconds was randomly implemented per request in addition to a 20-minute break between outlets. Despite these efforts, retrieval was still affected for several outlets, especially those behind paywalls. This resulted in uneven article counts across outlets and regional groups.

The first stage of article collection resulted in outlet-level csv files containing scraped articles from designated outlets. The following stage removed duplicate articles based on URL matches, and applied a keyword verification mechanism to exclude articles retrieved by GNews that didn’t directly relate to Greenland. Keywords included: "greenland", "greenlandic", "greenland’s", "greenlander", "greenlanders", "greenlander’s" and "nuuk". Remaining articles were sorted into csv files based on regional grouping. The final stage implemented the newspaper3k library to extract full article body texts with multi-threading (10 article downloads at a time) which removed those articles for which text extraction failed and those with less than 100 characters. This resulted in csv files representing each group containing articles based on regional position and their full texts. A cleaning filter was subsequently applied by manually identifying noisy

Table 1: Overview of the reduction in dataset size throughout the 3-stage verification process to eliminate noisy articles irrelevant to the Greenland crisis

Region	Stage 1 (query)	Stage 2 (verification)	Stage 3 (extraction)
Greenland & Denmark	1475	239	137
Affected Europe	7616	869	736
USA	6511	1075	533
Other	5810	901	649
Total	21412	3084	2055

data. This eliminated texts that alluded to common advertising or website statements such as subscription alerts or social media prompts.

The percentage retained after stage 2 suggests that the query-based search in stage 1 was not sufficient in solely extracting the relevant articles. Notably, the drop in article amount for region USA from stage 2 to 3 indicates the recurrent use of paywalls when attempting to access article texts.

4.3 *Handling data imbalance*

The significant under-sampling of region Greenland & Denmark signifies their lack of English-language coverage in news media. However, due to this group’s relevance to the crisis, a stratified down-sampling process was undertaken on the remaining groups to handle this major data imbalance. In order to maintain representational balance within each down-sampled regional group, different measures were taken: proportional per political affiliation for group USA (left, left-leaning, center, right leaning and right outlets), and down-sampled by country for groups Affected Europe and Other. This resulted in a final dataset with 137 articles for Greenland & Denmark, 595 for Affected Europe, 500 for USA and 442 for Other, with a total of 1,674 articles.

Strikingly, article counts across outlets vary substantially due to local access limitations and retrieval successes, constituting a limitation of the dataset to be considered throughout this thesis. A clear example of this is outlet Arctic Today ($n = 140$) and CNN ($n = 118$) dominating the dataset compared to The Spectator with only 2 articles or Spiegel with 1 retrieved article.

4.4 *Topic modeling*

Due to the absence of pre-existing thematic labels in the dataset, supervised classification becomes infeasible. A topic modeling technique that suffices

Table 2: Coherence and logged perplexity scores after LDA modeling for a range between 4-10 topics (k)

k	Coherence	Perplexity
4	0.4331	-7.4863
5	0.4532	-7.4887
6	0.4166	-7.4852
7	0.4072	-7.4823
8	0.3923	-7.4861
9	0.3913	-7.4870
10	0.4076	-7.4826

is Latent Dirichlet Allocation (LDA), known for its unsupervised, hierarchical nature. As introduced by [Blei, Ng, and Jordan \(2003\)](#), LDA is a Bayesian model that represents corpora as a combination of latent topics formed by a probability distribution over words. In this way, recurrent themes that are significant across corpora are quantitatively brought forth in the absence of predetermined topic labels. LDA has been used extensively in topic modeling studies to reveal thematic differences between corpora, such as the work by [Wang and Yin \(2022\)](#), where topical differences between two geopolitical groups narrating the same event were compared against each other.

A number of pre-processing steps were executed prior to modeling, using spaCy's `en_core_web_sm` pipeline. This is an essential step to enhance the model's ability to discriminate topics. Lowercasing, tokenization, removal of stop words using spaCy's English stopword list, non-alphabetic tokens and words shorter than three characters, and lemmatization was conducted. Hence, articles were reduced to space-separated lemmas.

A hyperparameter k number of topics must be predefined, which following the work by [Wang and Yin \(2022\)](#), can be deduced by calculating the coherence and perplexity scores within a range of k values. Perplexity deduces how well the model predicts the data while coherence infers each topics' semantic similarity of top words. An optimal k value can be found at the maximum C_v coherence and lowest log-likelihood perplexity scores. As seen in table 2, an optimal $k=5$ topics was found, yielding a coherence of 0.4532 and perplexity of -7.4887.

LDA utilizes the `gensim` library to create a dictionary of every token in the preprocessed text file and convert into a bag-of-words. Thus, a general and per region LDA model was fitted on the full corpus ($n=1,674$) with $k=5$ and `passes=20`. Under the general model, this resulted in the following five topics:

- Topic 1: 0.022*"greenland" + 0.014*"trump" + 0.013*"say" + 0.010*"party" + 0.010*"independence" + 0.009*"election" + 0.008*"denmark" + 0.006*"greenlandic" + 0.006*"people" + 0.006*"nuuk"
- Topic 2: 0.019*"greenland" + 0.007*"company" + 0.007*"year" + 0.005*"say" + 0.005*"project" + 0.004*"market" + 0.004*"new" + 0.004*"ice" + 0.004*"mining" + 0.004*"earth"
- Topic 3: 0.032*"trump" + 0.019*"greenland" + 0.015*"say" + 0.012*"president" + 0.008*"tariff" + 0.007*"european" + 0.006*"nato" + 0.006*"world" + 0.006*"denmark" + 0.005*"united"
- Topic 4: 0.047*"greenland" + 0.020*"trump" + 0.016*"say" + 0.016*"denmark" + 0.009*"president" + 0.009*"danish" + 0.009*"arctic" + 0.008*"island" + 0.008*"security" + 0.007*"visit"
- Topic 5: 0.040*"greenland" + 0.024*"say" + 0.020*"denmark" + 0.018*"trump" + 0.013*"nato" + 0.011*"danish" + 0.010*"arctic" + 0.010*"military" + 0.010*"minister" + 0.008*"president"

Inspection of the 80 most frequent terms under Python's Collections package across the dataset, found that top words such as "greenland" with a count of 17,272, "trump" with 10,243 counts, "say" with 8,571, and more (found in appendix A) appeared across topics and regional groups. These lack thematic relevance and disallow for the emergence of distinct clusters through LDA. Words from this list were manually removed before reapplying the LDA framework based on the researcher's judgment of its usefulness in topic modeling, such as "greenland", "president" and "country" that yield very little thematic relevance due to its broad scope within the subject of the crisis. Other words such as "include", "go", "come", "need" and "think" constitute as verbs and therefore hold minimal strength in topic modeling. Lastly, words such as "world", "year" and "post" serve as functional noise and were therefore also removed.

Subsequently, LDA was performed on the revised data accounting for global noise, following the same pipeline as earlier. A general LDA model was fitted on the full corpus to establish a standard framework for addressing RQ1.

To commence cross-regional topic modeling, topics under the general LDA results were manually assigned thematic labels as performed by [Wang and Yin \(2022\)](#) for an ease of understanding. LDA models were consequently fitted per regional group preserving the same parameters, yielding themes that were matched against the general thematic labels. This was done by computing the cosine similarity between the weight vectors of the top 10 words for each.

Prior to global noise removal, Principal Component Analysis as described by [Abdi and Williams \(2010\)](#) was applied to weighted word matrices per regional group. Each vector in this plot represents a topic formed from its top 10 word weights and is projected into a five dimensional principal component space. Each topical position measures where this topic sits relative to all other topics along both dimensions.

4.5 *Sentiment analysis*

Leveraging LDA per regional group elicited a clearer distinction between regions through their thematic choices when narrating the same event. In order to proceed with RQ2, this NLP framework shifts from topic modeling and PCA to sentiment analysis, building upon the work of [Gamea et al. \(2026\)](#).

The implemented pipeline for this section is as follows: a baseline model to assert a foundational measure of cross-regional emotional tone, and later expand on with an advanced model. Hence, Valence Aware Dictionary and sEntiment Reasoner was applied as a baseline for its ability to assign compound scores (ranging between -1 and 1) to corpora without the need for pre-processing. The underlying mechanism that drives these scores is a curated dictionary of words that VADER relies on, bearing sentiment and weighted intensity. A score is classified as negative (between -1 and -0.05), neutral (between -0.05 and 0.05) or positive (0.05 to 1). Hence, the `SentimentIntensityAnalyzer` class in the `vaderSentiment` library was applied. Previous work by [Thomsen \(2018\)](#) validates VADER’s effectiveness on corpora such as social media posts, and is relatively unpopular on full article texts. This motivates the choice of harnessing VADER as a baseline, and to compare to what extent subtle nuances are present in news media that may be captured more coherently when transitioning to transformer-based models.

To assess the computed compound scores for each article, a Kruskal-Wallis H-test is used to compare scores as it is able to handle VADER’s bounded scores between -1 and 1 where ANOVA can’t. This statistical test ranks all scores in ascending order and deduces their significance. In turn, pairwise comparisons were conducted across each possible pairs of regions (6 in total) with Mann-Whitney U tests and following the work of [Cheng, Vasan, Patel, and Shekane \(2025\)](#), Bonferonni correction was applied to mitigate the possibility of finding a significant result by chance:

$$\alpha_{\text{corrected}} = \frac{0.05}{6} = 0.0083 \quad (1)$$

Moving beyond the baseline as executed by [Gamea et al. \(2026\)](#), a RoBERTa (Robustly Optimized BERT Approach) model is implemented, a transformed-based language model and extension of BERT that makes use of dynamic masking and large batch training times to strengthen contextual representations. Whereas VADER scores based on individual words' sentiment, RoBERTa encodes tokens based on full context. ? found that this model is proficient in exploring sentiment in complex political language, taking into account the use of sarcasm, formal language use and more.

With a maximum of 512 input tokens, texts were split into 400 word chunks per article when necessary to account for this. Chunks were labeled positive, neutral or negative given the assigned confidence score which was converted to a -1 to 1 scale to align with VADER's scoring system. For each article, these scores per chunk were averaged to represent its overall sentiment.

The [siebert/sentiment-roberta-large-english](#) RoBERTa model of [Hartmann, Heitmann, Siebert, and Schamp \(2023\)](#) was leveraged due to its training on a diverse range of English corpora relevant to this thesis, including formal language domains such as news articles.

Hence, the same statistical tests were applied on results as with VADER, and Pearson's correlation coefficient was calculated between VADER and RoBERTa scores per region to assess both models against each other.

5 RESULTS

5.1 *Topic Modeling with LDA*

The following topics are the result of manually assigning thematic labels on general LDA results based on qualitative analysis of the top words per topic:

- Topic 1 - Geopolitical sovereignty & security: 0.011*"military" + 0.011*"nato" + 0.010*"minister" + 0.010*"arctic" + 0.010*"security" + 0.008*"island" + 0.008*"territory" + 0.006*"prime" + 0.006*"ally" + 0.005*"foreign"
- Topic 2 - Natural resources & global interests: 0.011*"company" + 0.008*"project" + 0.005*"mineral" + 0.005*"mining" + 0.004*"market" + 0.004*"rare" + 0.004*"molybdenum" + 0.004*"development" + 0.004*"million" + 0.004*"earth"
- Topic 3 - Geographic & strategic positioning: 0.011*"arctic" + 0.007*"island" + 0.006*"ice" + 0.004*"base" + 0.004*"china" + 0.004*"military" + 0.004*"nuuk" + 0.004*"north" + 0.003*"mineral" + 0.003*"resource"

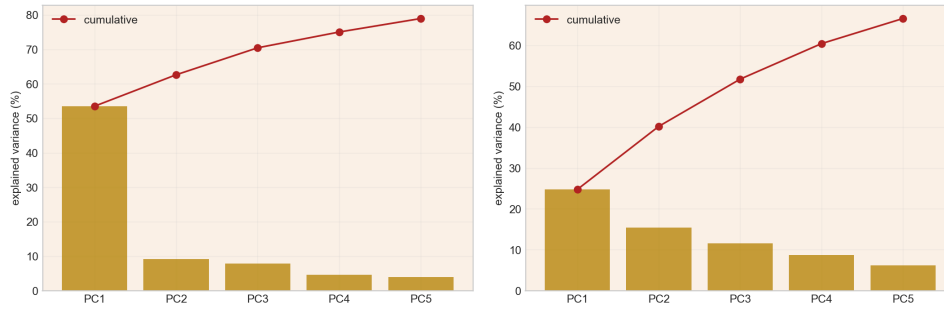


Figure 1: Scree plot of LDA results across regions depicting the explained variance for five principal components before (left) versus after (right) global noise removal

- Topic 4 - Economic pressure & global relations: $0.010 \times \text{"european"} + 0.010 \times \text{"tariff"} + 0.007 \times \text{"nato"} + 0.007 \times \text{"europe"} + 0.005 \times \text{"force"} + 0.005 \times \text{"deal"} + 0.005 \times \text{"threat"} + 0.005 \times \text{"davos"} + 0.005 \times \text{"ally"} + 0.005 \times \text{"economic"}$
- Topic 5 - Diplomatic engagement: $0.015 \times \text{"visit"} + 0.014 \times \text{"vance"} + 0.009 \times \text{"security"} + 0.006 \times \text{"base"} + 0.006 \times \text{"tariff"} + 0.005 \times \text{"vice"} + 0.005 \times \text{"minister"} + 0.005 \times \text{"government"} + 0.005 \times \text{"territory"} + 0.004 \times \text{"usha"}$

The best match results for each topic per region can be found in appendix C. Strikingly, regional group Greenland & Denmark maps four of its five topics to Geopolitical Sovereignty & Security, suggesting a recurring narrative focused on independence.

The scree plot in figure 1 before noise removal shows that PC1 explained 53.5% of variance, predominantly loaded by generic crisis-related vocabulary such as "greenland" (0.625), "trump" (0.512) and "say" (0.359). This suggests that high-frequency words governed the feature space, causing regions to be discriminated by their frequency across this generic vocabulary rather than thematic content. Consequently, topics weren't distinguishable.

Accordingly, a PCA plot that implemented the exclusion of global noise resulted in a substantial redistribution of variance across components as seen in figure 1 (PC1 = 24.8%, PC2 = 15.4%, PC3 = 11.6%). Principal components loaded on words more interpretative to thematic constructs than with global noise. Figure 2 presents a PCA plot depicting each theme per region and its position on the PC1 and PC2 axes accompanied by a bi-plot showcasing the dominant loading words and their relative directions on the PCA plot. The positive PC1 area of the plot loaded words such as "security", "nato" and "military", and the negative side loaded words such as "market", "earth" and "mineral". PC2 captured the major diplomatic event ("visit", "vance", "independence") loading on its positive area of the

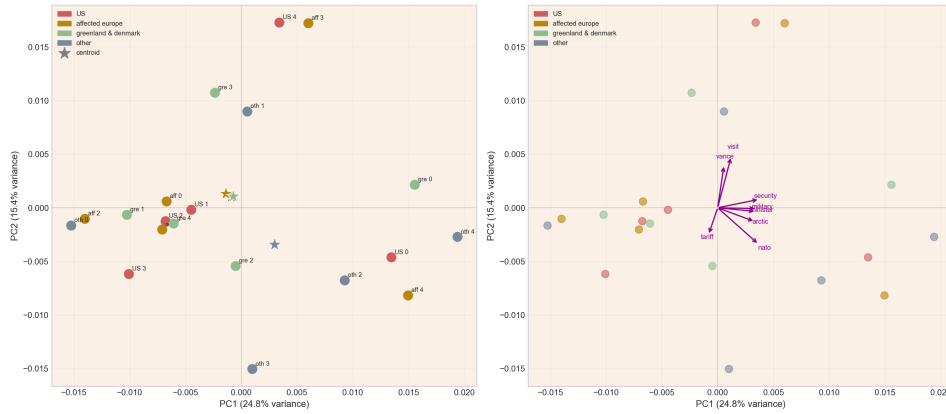


Figure 2: PCA plot for two principal components of LDA topic distributions across regions (left) & bi-plot containing arrows of the highest frequency word loadings and their designated directions on the PCA plot (right)



Figure 3: Word clouds depicting the most frequent words for groups USA topic 4 and Affected Europe topic 4, both falling under the LDA match: Geopolitical sovereignty & security

plot and "arctic", "ally" and "tariff" residing on the negative. Centroids showcase the mean position over every topic per region. Regional group Other deviates from the other centroids that cluster near the origin, residing towards the negative PC1 pole.

Figure 3 presents word clouds of the dominant vocabulary in USA & Affected Europe's topic 4 that both fall under the same thematic label of Geopolitical Security & Sovereignty. The US topic is dominated by words such as "visit", "vance", "military" and "island". The latter includes words such as "arctic", "military", "nato" and "security".

5.2 Baseline Sentiment Analysis with VADER

Table 3 presents VADER results per region, with average compound scores, standard deviations and the percentage of negative and positive articles contained. All four regions received a positive sentiment score, with

Table 3: Statistical analysis of VADER compound scores per region with the amount (%) of negative versus positive articles

Region	Mean	SD	Negative %	Positive %
Greenland & Denmark	0.67	0.56	10.9	86.1
Affected Europe	0.59	0.62	16.8	80.7
USA	0.52	0.72	21.8	77
Other	0.42	0.73	26.7	70.4

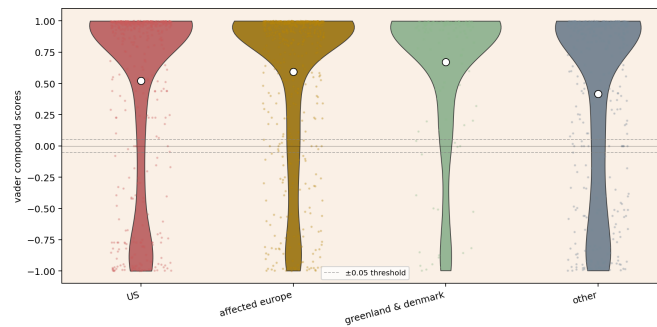


Figure 4: Violin plot for VADER compound score results per region (white dot indicates the mean)

region Greenland & Denmark having the most positive overall tone (mean of 0.67) and region Other the least (mean 0.42). Standard deviations per regional group ranged between 0.56 to 0.73. A Kruskal-Wallis test indicated statistically significant general differences between regional distributions ($H=16.95$ and $p<0.05$). As seen in appendix D, a pairwise Mann-Whitney U test with Bonferonni correction linked region Other with p-values below 0.05 when assessed against the three stakeholder regions whereas pairs excluding Other resulted in p-values above 0.4.

As utilized by ?, a violin plot was created to visualize the results is shown in figure 4, with all four distributions being positively skewed and residing near the 1.00 score mark. Region Other is slightly differently skewed from the remaining regions, as supported by table 3's results.

5.3 Advanced Modeling: Score Distributions

Table 4 presents RoBERTa modeling results per region, with mean scores ranging between 0.04 (group US) and 0.39 (Affected Europe). A Kruskal-Wallis test was highly significant ($H=76.72$, $p<0.0001$). Furthermore, as seen in appendix E, the pairwise Mann-Whitney U with Bonferonni correction as seen in Appendix E indicated that aside from the Greenland & Denmark

Table 4: Statistical analysis of RoBERTa sentiment scores per region with the amount (%) of negative versus positive articles

Region	Mean	SD	Negative %	Positive %
Greenland & Denmark	0.26	0.84	28.5	54
Affected Europe	0.39	0.81	26.2	65.9
USA	0.04	0.81	38.6	42.6
Other	0.2	0.81	32.6	53.4

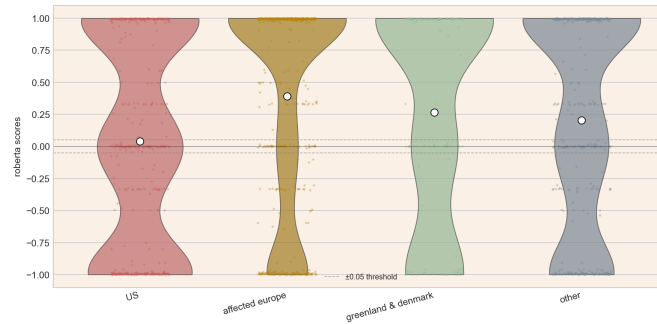


Figure 5: Violin plot for RoBERTa sentiment score results per region (white dot indicates the mean)

and Other pair ($p=0.158$), the 5 other pairs were significant. Standard deviations per region ranged between 0.81 to 0.84.

Figure 5 presents a violin plot for RoBERTa scores, depicting distributional differences per regional group. Group US's shape is notably wider around a score of 0 and at its tail compared to the other regions, reflecting table 4's results. Affected Europe has the most positively skewed shape with little width in the middle, and Greenland & Denmark and Other are similar in its positively skewed shape and its width around the middle being between that of USA and Affected Europe.

5.4 Advanced modeling: Qualitative Evaluation

To understand how RoBERTa captures tonal differences and distributes sentiment throughout the dataset, intriguing cases of article scores are examined by the chunks they are comprised of.

Under USA, the most negatively scored article (-0.999) is a film review of Greenland 2: Migration' review: The destruction of Earth is the least of Gerard Butler's worries by the New York Post. Critical language is used, such as "snoozy" and "awfully". This article refers to a film rather than the Greenland crisis, revealing the presence of topical noise in the dataset. The most positively scored text (0.999) in this region by the New York Times

consists of brief texts of speech quotations that were exchanged during a press conference. This suggests the inflation of positive scores when language use is sparse and dominated by quotes.

One of region Greenland & Denmark's most positively scored (0.999) article reports a diplomatic interaction between two countries in solidarity with each other. Consisting of neutral and formal words, this suggests that RoBERTa was able to pick up on the contextual significance of this text. Conversely, the most negatively scored article (-0.999) opens with a rejection and defensive language, such as the chunk: "I am strongly opposed to this appointment and especially the statements made by the special emissary, which I find totally unacceptable". RoBERTa picked up on this defensive language and correctly interpreted its tone. Lastly, the most neutrally scored article (0.000) touches on a cancellation of King Frederik's visit to Greenland. The following excerpts within this article represent the most negatively and positively scored chunks:

- Chunk 1 (-0.999): "King Frederik's visit to Greenland has been disrupted by weather conditions, with several cancellations and other changes to the original schedule. A planned visit on Wednesday by King Frederik and..."
- Chunk 2 (0.999): "my connection with the Greenlandic people is intact."

This illustrates how chunk averaging is processed, with mixed scores at extremes producing an average score of 0. Here, RoBERTa focuses on individual passages rather than topics in the full text which is result of its 512-token limit.

5.5 *Comparison between Baseline vs Advanced modeling*

Figure 6 reveals the difference in scoring between the baseline and advanced models. VADER scored the texts much more positively than RoBERTa did and implied very little differences in sentiment between regions. On the other hand, RoBERTa scores deviate remarkably. The Pearson's Correlation Coefficient supports this with $r=0.264$, suggesting a very weak correlation between both models. Per-region correlations further suggested this with statistically significant results between 0.17 and 0.3.

6 DISCUSSION

The goal of this thesis was to explore how natural language processing tools can aid in detecting the influence of geopolitical position on news media

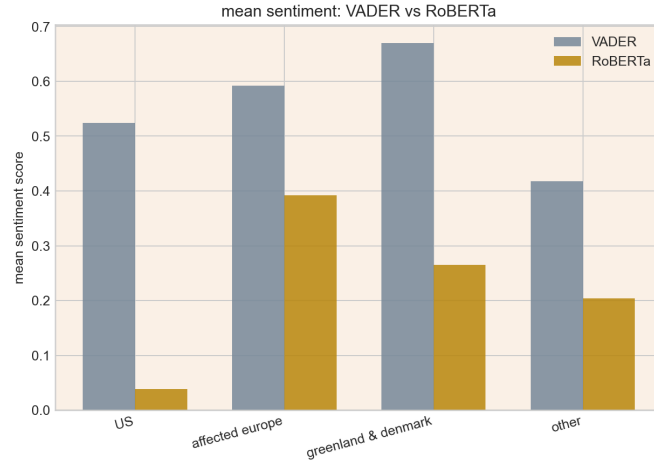


Figure 6: Mean sentiment score comparison between VADER & RoBERTa

narratives covering the same global event. Two approaches were taken; topic modeling with LDA and sentiment analysis with baseline model VADER and advanced model RoBERTa. These approaches further explored three sub-questions regarding the dominating themes, emotional tone of stakeholder regional news media outlets, and alignment or divergence of the narratives of non-affected regions. Results indicate geopolitical position as an influential factor shaping the framing and sentiment of news media. Although, by transitioning from a traditional to transformer-based model, it is apparent that this relationship is nuanced and easily overlooked.

6.1 Dataset Collection

The process of collecting articles from different regional outlets brought forth some limitations in data collection. This thesis was bounded to articles written in English, and considering the importance of gathering global coverage and narratives, this restriction may have caused an under-representation of domestic regional perspectives due to the vast availability of articles for English-speaking regions and the limited amount for the latter. Greenland in the Greenland & Denmark group was severely under-represented due to the lack of English-written articles. However, this region was preserved because it served as a significant stakeholder group, and can be as thematically dense as larger sized groups. Moreover, while attempts were made to maintain a balance, an imbalance was caused in articles on an outlet-level. ArcticToday dominated the dataset with 140 articles, while others such as Spiegel contained just 1.

Furthermore, the 3-stage data collection process focused on a keyword-based filtering of articles with generic words such as "greenland", therefore not effectively eliminating irrelevant articles. This includes those articles where the keyword is present but are irrelevant to the crisis itself. This can have had an effect on results as observed in the qualitative analysis, however through topic modeling this issue isn't as severe.

6.2 *Topic Modeling: LDA distributions*

Appendix C denotes the best matches between each topic and the constructed thematic labels per region along with their scores. Results indicate Greenland & Denmark's thematic focus on topics underlying sovereignty and security while remaining regions retain a more disperse thematic scope. This finding appropriately demonstrates how a primary and threatened stakeholder group may adopt narratives that deviate from global viewpoints and align with national interests. It is crucial to highlight that while the thematic labels serve as a helpful analytical tool to interpret results, labels were finalized based on the researcher's own interpretation. They do not reflect the ground truth, which constitutes a known limitation of unsupervised topic modeling.

The significant redistribution of results after removing noisy and frequent terms confirms the necessity of excluding generic event-related vocabulary from LDA modeling to enhance topic clusters. Examining individual topics for regional group Other; topic 4 of figure 2 occupies the most positive PC1 position. On the opposite side of the plot along the PC1 plot lies topic 0, and topic 3 is positioned most negatively on PC2 and almost centrally on PC1. Its centroid deviates greatly from those of the other groups, inferring that non-affected regional outlets employ heterogeneous vocabulary when narrating this crisis under similar themes as affected regions.

A concentration of topics residing left on PC1 and central on PC2 is found, revealing thematic overlap across regions (topics 1 and 2 for US, topics 0 and 1 for Affected Europe and topic 4 for Greenland & Denmark). A handful of vocabulary is used across regions under different themes, indicating that despite the removal of global noise, much vocabulary is shared that doesn't strongly refer to one thematic frame.

Where the LDA matches were found by calculating the cosine similarity between general and region-specific topical word weights, the positioning of topic plots in PCA is relative to all other topics along the PC dimensions. For example, topic 4 under regional group USA and topic 4 under Affected Europe both match with the general topic Geopolitical Security & Sovereignty, however on the PCA plot they sit at very different positions.

The former resides extremely positively on PC2 and almost central on PC1, while the latter resides by the bottom right corner of the plot. Differences in internal vocabulary of both topics that converge on the theme of Geopolitical Security & Sovereignty is captured by PCA, demonstrating how vocabulary can reveal subtle narrative divergence under the same theme. Word clouds in figure 3 where both regions' topics fall under the thematic label of Geopolitical Security & Sovereignty, insinuate this. US topic 4 alludes to a specific event, namely the diplomatic visit by JD Vance of Greenland. On the other hand, Affected Europe topic 4 indicates a more widespread international alliance and threat framework. These different angles illustrate how geographical position influences subtle topical standpoints.

6.3 *Sentiment analysis: baseline score distributions*

The lack of differentiability in scores between regions suggest that VADER is unable to detect tonal nuances in texts that aren't explicitly expressive. VADER was trained on social media corpora and past research, hence, work by [Ribeiro, Araújo, Gonçalves, André Gonçalves, and Benevenuto \(2016\)](#) has shown that it under-performs on formal texts. Furthermore, scores are consistently positive among regions, suggesting that VADER misses tonal shifts in news media. This implies that lexicon-based models are inadequate in detecting sentiment in such corpora, supporting [Gamea et al. \(2026\)](#)'s methodology of building upon the baseline VADER model.

A noteworthy observation: region Other slightly deviates from the remaining regions in compound score, with the least positive scores (70.4% positive) as seen in figure 3. This slight but apparent shift in sentiment indicates a less positive tone to the crisis when being narrated from an outsider's perspective.

6.4 *Sentiment analysis: advanced score distributions*

The Kruskal-Wallis test revealed a H statistic 4 times larger for RoBERTa than that of VADER, a direct indication of an advanced models' ability to find sharper distinctions in sentiment across regions. Additionally, the Mann-Whitney U tests revealed that region Other is not an outlier. Instead, it is nonsignificant with Greenland & Denmark ($p = 158$). The relationship between groups with opposite proximity to the crisis, sharing the same emotional tone (see figure 5), does not appear linear. These groups fall at opposite ends of affiliation levels to the crisis and diverge from regions at intermediate levels, suggesting that proximity to the crisis alone is not an appropriate predictor of sentiment.

While transitioning to the RoBERTa model resulted in greater regional differences, due to the selection of the SiEBERT model without fine-tuning, the level of contextual understanding the model harnessed was limited. An improvement is to incorporate fine-tuning techniques within the methodology to leverage a greater understanding of topic-specific vocabulary used in the dataset.

6.5 *Sentiment analysis: Qualitative Evaluation of RoBERTa*

Addressing the cross-regional article exemplars of RoBERTa's sentiment scores; the negatively-scored article under USA constitutes an important limitation of the data collection process as it entails a subject outside of the Greenland crisis. This is a result of the keyword-based search for articles used when building the dataset, whereby articles that do not relate to the crisis but fall under the domain of 'Greenland' are captured.

The chunk-averaging process applied to tackle RoBERTa's 512-token limit caused the article under Greenland & Denmark to be scored neutral despite the extremely polarized chunks. This indicates how chunk-averaging can result in misleading sentiment scores and suggests the implementation of a manual verification procedure for extreme scoring cases.

Aside from these limitations, as seen with the positively scored article under Greenland & Denmark, RoBERTa is sensitive to diplomatic and solidarity language. This implies the relevance of the SiEBERT model to handle this dataset as it yields a contextual understanding of political language. Furthermore, its most negatively scored article further validates this, as defensive and intensely-charged language was appropriately interpreted. These tonal differences were detected because context is encoded across full sentences rather than in individual words, which reinforces its extensive capabilities.

6.6 *Scientific & Societal Relevance*

The relevance of this thesis covers much ground. Firstly, the immense cross-regional shift in sentiment results when transitioning from a lexicon to transformer-based model following [Gamea et al. \(2026\)](#)'s approach contributes to empirical evidence of the development of NLP tools. Besides, highlighting how geopolitical position can shape media narratives can unveil the partial views of global events that audiences of news media consume, and encourage the awareness of such frames. This societal implication is crucial in considering global perspectives and supporting the critical analysis of media in any form.

To extend this research, a couple of directions can be followed: topic modeling can be delved into more thoroughly with a state-of-the-art model. As [Son and Park \(2025\)](#) deduces, BERTopic modeling can produce more semantically distinguishable topic clusters than traditional models can. Moreover, a multilingual analysis can be implemented to rule out the limitation of being bound to English articles causing an under-representation of some regions. Lastly, with a larger, balanced corpora, outlet-level sentiment analysis can be considered to examine whether outlets within regions moderate the resulting patterns observed in this thesis, for example, when considering political affiliation.

7 CONCLUSION

This thesis explored the extent of influence geopolitical position has on the narration of a global event by news media outlets. Three sub-questions were investigated through an NLP framework, combining topic modeling with LDA accompanied by PCA and sentiment analysis with VADER and RoBERTa's SiEBERT model.

LDA uncovered thematic mappings across the four regional groups, finding Greenland & Denmark to match closely to topics underlying geopolitical security & sovereignty. Topical distributions with PCA suggested that region Other obtained the most thematic heterogeneity, with topics positioned at extreme points on the PC1 and PC2 axes.

Moreover, VADER's known usage restricted to social media prose was evident in its scoring, finding most regions to exhibit a uniformly positive sentiment with a slight deviation by region Other. RoBERTa computed deviating scores, with much sharper regional differences. Region USA emerged as the most negatively scored, Affected Europe appeared the most positive and Greenland & Denmark and Other were practically indistinguishable. This similarity suggests that the level of involvement in the crisis and the emotional tones do not imply a linear relationship.

In conclusion, the aforementioned findings reinforce the influence geopolitical alignment leverages on news media narratives. By utilizing advanced NLP tools to tackle topic modeling and sentiment analysis, these nuances can be detected and justified in formal news prose.

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8 APPENDICES

APPENDIX A

Top 15 most frequent word occurrences in the dataset:

greenland: 17272
 trump: 10243
 say: 8571
 denmark: 6168
 president: 4250
 danish: 3433
 arctic: 3321
 security: 2958
 military: 2815
 united: 2763
 nato: 2699
 island: 2536

states: 2484
 minister: 2461
 territory: 2189

APPENDIX B

LDA results for k=5 topics without global noise per region

Region Greenland & Denmark:

- Topic 1: 0.013*"minister" + 0.010*"security" + 0.010*"territory" + 0.010*"arctic" + 0.009*"visit" + 0.009*"frederiksen" + 0.008*"prime" + 0.007*"nato" + 0.007*"military" + 0.006*"foreign"
- Topic 2: 0.006*"mineral" + 0.006*"lauder" + 0.005*"interest" + 0.005*"company" + 0.005*"water" + 0.005*"investment" + 0.004*"mining" + 0.004*"business" + 0.004*"write" + 0.004*"accord"
- Topic 3: 0.013*"rutte" + 0.009*"nato" + 0.008*"howery" + 0.006*"ambassador" + 0.004*"minister" + 0.004*"prime" + 0.004*"property" + 0.004*"dutch" + 0.004*"arctic" + 0.004*"buy"
- Topic 4: 0.014*"independence" + 0.011*"party" + 0.009*"election" + 0.009*"government" + 0.007*"greenlanders" + 0.007*"visit" + 0.006*"military" + 0.006*"arctic" + 0.006*"percent" + 0.006*"island"
- Topic 5: 0.008*"sheridan" + 0.008*"nathan" + 0.007*"arctic" + 0.005*"funding" + 0.005*"state" + 0.005*"ngos" + 0.004*"government" + 0.004*"king" + 0.004*"work" + 0.004*"aid"

Region USA

- Topic 1: 0.010*"nato" + 0.009*"military" + 0.009*"security" + 0.008*"minister" + 0.008*"island" + 0.007*"arctic" + 0.007*"territory" + 0.006*"european" + 0.006*"ally" + 0.006*"prime"
- Topic 2: 0.009*"island" + 0.008*"mineral" + 0.008*"arctic" + 0.007*"earth" + 0.006*"rare" + 0.006*"china" + 0.005*"military" + 0.005*"ice" + 0.005*"resource" + 0.004*"mining"

- Topic 3: 0.005*"american" + 0.005*"america" + 0.005*"deal" + 0.003*"state" + 0.003*"independence" + 0.003*"nuuk" + 0.003*"war" + 0.003*"force" + 0.003*"nato" + 0.003*"defense"
- Topic 4: 0.014*"tariff" + 0.010*"market" + 0.007*"bessent" + 0.005*"bond" + 0.005*"trade" + 0.004*"european" + 0.004*"day" + 0.004*"economic" + 0.004*"investor" + 0.004*"high"
- Topic 5: 0.016*"vance" + 0.013*"visit" + 0.011*"security" + 0.007*"base" + 0.006*"vice" + 0.006*"military" + 0.006*"space" + 0.005*"island" + 0.005*"trip" + 0.005*"house"

Region Affected Europe

- Topic 1: 0.006*"ice" + 0.004*"election" + 0.004*"arctic" + 0.003*"power" + 0.003*"party" + 0.003*"independence" + 0.003*"child" + 0.003*"nuuk" + 0.003*"long" + 0.003*"keira"
- Topic 2: 0.006*"tariff" + 0.005*"nuuk" + 0.005*"state" + 0.004*"island" + 0.004*"european" + 0.004*"american" + 0.003*"territory" + 0.003*"america" + 0.003*"government" + 0.003*"international"
- Topic 3: 0.011*"project" + 0.009*"company" + 0.008*"mining" + 0.006*"mineral" + 0.005*"resource" + 0.005*"rare" + 0.005*"gold" + 0.005*"earth" + 0.004*"government" + 0.004*"ice"
- Topic 4: 0.013*"visit" + 0.010*"vance" + 0.010*"minister" + 0.008*"island" + 0.008*"territory" + 0.008*"base" + 0.007*"prime" + 0.007*"security" + 0.007*"party" + 0.006*"independence"
- Topic 5: 0.016*"arctic" + 0.011*"nato" + 0.011*"military" + 0.009*"security" + 0.008*"minister" + 0.006*"ally" + 0.005*"european" + 0.005*"russia" + 0.005*"island" + 0.005*"force"

Region Other

- Topic 1: 0.014*"company" + 0.010*"molybdenum" + 0.009*"project" + 0.006*"earth" + 0.006*"look" + 0.006*"market" + 0.006*"magnesium" + 0.006*"rare" + 0.005*"mineral" + 0.005*"statement"

- Topic 2: 0.008*"visit" + 0.007*"island" + 0.007*"vance" + 0.005*"security" + 0.005*"territory" + 0.005*"market" + 0.004*"military" + 0.004*"independence" + 0.004*"arctic" + 0.004*"interest"
- Topic 3: 0.014*"nato" + 0.009*"military" + 0.009*"arctic" + 0.006*"force" + 0.006*"security" + 0.006*"russia" + 0.006*"defence" + 0.005*"base" + 0.005*"ally" + 0.004*"territory"
- Topic 4: 0.015*"tariff" + 0.015*"european" + 0.009*"nato" + 0.008*"threat" + 0.008*"ally" + 0.007*"europe" + 0.007*"canada" + 0.006*"leader" + 0.006*"security" + 0.006*"minister"
- Topic 5: 0.013*"military" + 0.012*"arctic" + 0.012*"nato" + 0.011*"security" + 0.011*"minister" + 0.009*"island" + 0.008*"territory" + 0.006*"prime" + 0.006*"house" + 0.006*"national"

APPENDIX C

Best matches between LDA results per region and general thematic labels (without global noise)

Region	Topic	Best match	Score
Greenland & Denmark	1	Geopolitical sovereignty & security	0.9598
	2	Natural resources & global interests	0.7978
	3	Geopolitical sovereignty & security	0.7978
	4	Geopolitical sovereignty & security	0.8075
	5	Geopolitical sovereignty & security	0.7548
Affected Europe	1	Geographic & strategic positioning	0.6506
	2	Economic pressure & global relations	0.8206
	3	Natural resources & global interests	0.8958
	4	Geopolitical sovereignty & security	0.8458
	5	Geopolitical sovereignty & security	0.9195
USA	1	Geopolitical sovereignty & security	0.979
	2	Geographic & strategic positioning	0.8206
	3	Economic pressure & global relations	0.8958
	4	Natural resources & global interests	0.8458
	5	Geopolitical sovereignty & security	0.9195
Other	1	Natural resources & global interests	0.9078
	2	Geopolitical sovereignty & security	0.8832
	3	Geopolitical sovereignty & security	0.8948
	4	Economic pressure & global relations	0.9312
	5	Geopolitical sovereignty & security	0.9646

APPENDIX D

Pairwise Mann-Whitney U with Bonferroni correction for cross-regional VADER compound scores

Pair	p-value
Greenland & Denmark vs Affected Europe	0.4033
Greenland & Denmark vs USA	0.9680
Greenland & Denmark vs Other	0.0043
Affected Europe vs USA	0.4388
Affected Europe vs Other	0.0010
USA vs Other	0.0005

APPENDIX E

Pairwise Mann-Whitney U with Bonferroni correction for cross-regional RoBERTa sentiment scores

Pair	p-value
Greenland & Denmark vs Affected Europe	0.0481
Greenland & Denmark vs USA	0.0003
Greenland & Denmark vs Other	0.1575
Affected Europe vs USA	0.0000
Affected Europe vs Other	0.0000
USA vs Other	0.0008